NeuroScan AI: Technical Summary

Explainable, agentic clinical decision support for brain MRI. B.Tech CSE mini project, Vaagdevi College of Engineering, Warangal. Guide: Dr. P. Shailaja. Author: Nikhil Chary Sriramoju.

Research prototype. Not a medical device. Not for clinical use.

1. Pipeline

Stage	What happens
Upload	PNG/JPEG only, content-sniffed by magic bytes, 8 MB cap
Guardrails	Non-MRI / out-of-distribution gate, PII and prompt-injection filtering
CNN	ResNet-18, 3 classes (Alzheimer's Disease, Brain Tumor, Normal Anatomy), temperature-scaled confidence
Grad-CAM	Heat map from the last conv block; confident-but-diffuse maps flagged
RAG	BM25 retrieval over ingested PDFs; extractive, cited passages only
Agents	Orchestrator with per-agent tool allow-lists; no agent can release a result
Clinician review	Mandatory human sign-off; only this step releases a case

2. Security design

PBKDF2 password hashing; HS256 JWT; role-based access (patient, clinician, admin); rate limiting; append-only hash-chained audit log verifiable at /api/audit/verify; Docker with non-root user, CSP headers, cap_drop ALL and a read-only filesystem. Backend core logic is covered by 27 automated tests run in CI.

3. Evaluation

Results are produced by `python -m ml.evaluate` on a held-out test split and written to `src/metrics.json`; the website renders them automatically. **No accuracy figure is claimed in this document** because the model has not yet been trained and evaluated. Metrics reported will include accuracy, per-class precision/recall/F1, confusion matrix and expected calibration error (ECE).

4. Limitations

Small public datasets; 2D slices only; three classes; Grad-CAM is not causal proof; the OOD gate is a heuristic; BM25 is lexical (no embeddings); SQLite and in-memory rate limits suit one worker; no clinical validation; no independent security audit.

5. Future work

Embedding-based retrieval, 3D volumes (DICOM/NIfTI), more disease classes, external validation, Postgres and Redis, SSO.

6. Links

Repository: github.com/Nikhil-creat/NeuroScan_AI-mini-project-
Live site: nikhil-creat.github.io/NeuroScan_AI-mini-project-/